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TrustGate: Stateful Runtime Authorization for Embodied AI Action Streams

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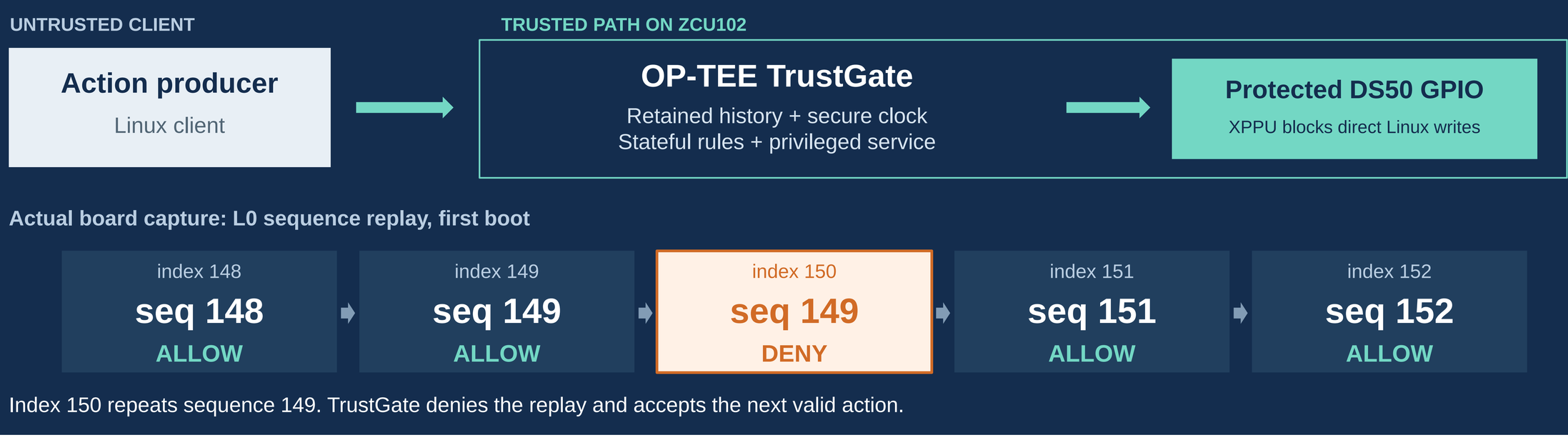
Abstract

Robot action packets may satisfy individual limits while violating stream rules. TrustGate places a stateful authorization monitor in OP-TEE on a Zynq UltraScale+ MPSoC. It retains history across client sessions within a boot, uses secure time, and mediates an XPPU-protected GPIO. We tested five action traces with five predeclared mutation classes. All 25 mutation points were denied on each of three board boots. Four encoded clean streams were fully accepted, while the fifth incurred 35 denials among 270 actions. A host-stateful baseline also detected all mutations on raw inputs. In 10,000 authorized-IDLE measurements, paired TIME and SUBMIT calls had a host-observed p99 of 4.75 ms. These results demonstrate trusted enforcement at the board GPIO. The robot servo bus remains outside the measured boundary.

1 Methods and replay evidence

Five traces, five mutation classes: replay, reorder, stale time, joint jump and forbidden transition.

25/25
mutation cases blocked on each of three boots

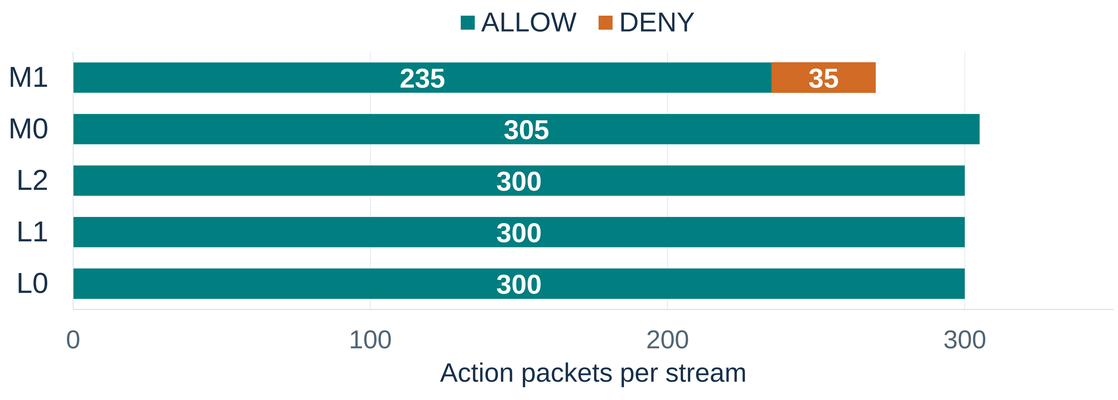


Index 150 repeats sequence 149. TrustGate denies the replay and accepts the next valid action.

2 Results

Four clean streams fully accepted

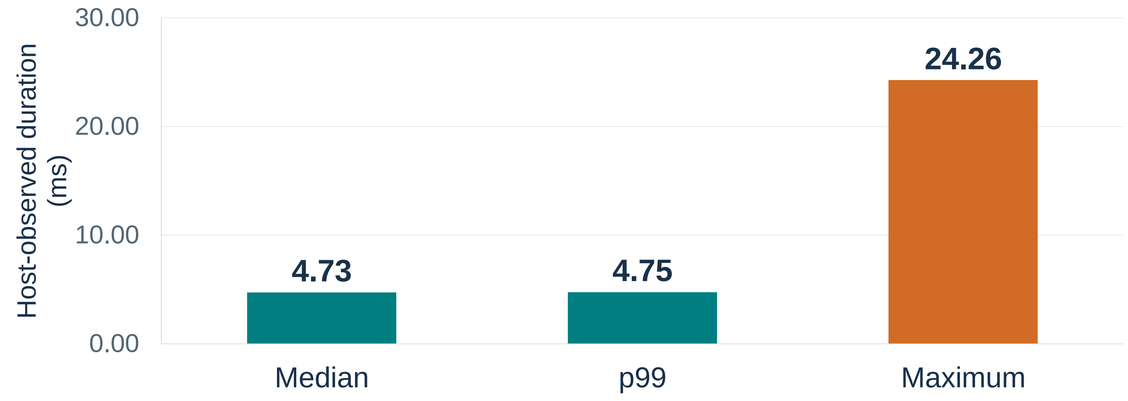
Encoded clean inputs, per board boot



L0-L2: SO-101 recordings. M0-M1: SmolVLA episodes. Counts repeat on three boots. M1: 5 joint-limit denials and 30 subsequent chunk-gap denials.

Service latency: p99 4.75 ms

10,000 paired TIME + SUBMIT calls



Maximum: 24.26 ms. One sustained authorized-IDLE boot, including UART diagnostics. PC transport and servo movement are excluded.

3 Conclusions and limitations

Trusted enforcement

A stateful host baseline detects the same mutations on raw inputs. TrustGate enforces these rules inside the trusted path.

Measured boundary

Protection ends at DS50 GPIO. The Windows PC still controls the SO-101 servo bus.

Next step

Protect the actuator bus, then measure end-to-end timing under robot workloads.

